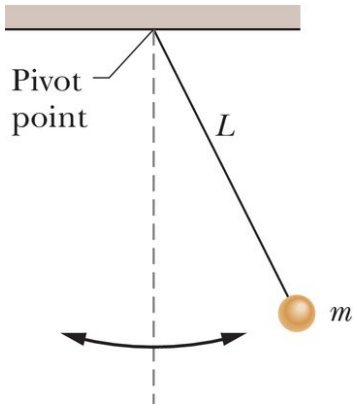
Lesson 15B Pendulums, Circular Motion

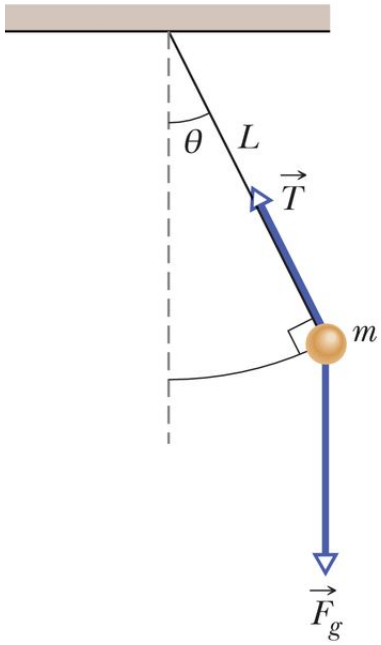
Pendulums and Simple Harmonic Motion

Simple Pendulum

A **simple pendulum** consists of a mass m (called the **bob**) attached to a massless, unstretchable string of length L. When displaced and released, the bob swings back and forth under the influence of gravity.



A free body diagram for the pendulum is shown in the following figure:



When the pendulum is displaced by a small angle θ, gravity creates a **restoring torque**:

$$\tau = -mgL\sin\theta$$

For small angles ($\theta \lesssim 15^\circ$), we can use the approximation $\sin\theta \approx \theta$, giving:

$$\tau \approx -mgL\theta$$

Using the rotational form of Newton's second law, $\tau = I\alpha$, and for a point mass at distance L, $I = mL^2$, we get:

$$mL^2\alpha = -mgL\theta \Rightarrow \alpha = -\frac{g}{L}\theta$$

This is the equation for **angular simple harmonic motion**, where angular acceleration is proportional to angular position and opposite in direction.

- The **angular frequency** is:

$$\omega = \sqrt{\frac{g}{L}}$$

- The **period** of the pendulum is:

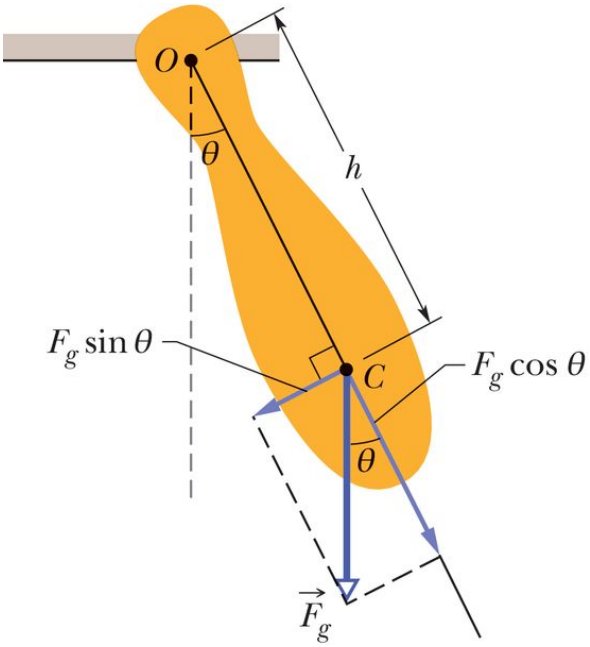
$$T = 2\pi\sqrt{\frac{L}{g}}$$

This result is valid only for small angular amplitudes, where the restoring torque is approximately linear in θ.

Pendulum animation

Physical Pendulum

A **physical pendulum** is any rigid object that swings about a pivot point. Unlike the simple pendulum, the mass is not concentrated at a point.



Let:

- I be the moment of inertia about the pivot
- m be the total mass
- h be the distance from the pivot to the center of mass

The torque is:

$$\tau = -mgh\theta$$

And using $\tau = I\alpha$, we get:

$$\alpha = -\frac{mgh}{I}\theta$$

This again leads to simple harmonic motion (SHM) for small θ, with:

- **Angular frequency**:

$$\omega = \sqrt{\frac{mgh}{I}}$$

- **Period**:

$$T = 2\pi\sqrt{\frac{I}{mgh}}$$

Notes:

- A physical pendulum will **not** oscillate in SHM if it's pivoted at its **center of mass**.
- The **center of oscillation** is the point where a **simple pendulum** of length L_0 would have the same period as the physical pendulum.

Using a Physical Pendulum to Measure g

A **uniform rod** of length L, pivoted at one end, is a common example. For this system:

- Moment of inertia: $I = \frac{1}{3}mL^2$
- Distance to center of mass: $h = \frac{L}{2}$

Plugging into the period equation:

$$T = 2\pi\sqrt{\frac{\frac{1}{3}mL^2}{mg \cdot \frac{L}{2}}} = 2\pi\sqrt{\frac{2L}{3g}}$$

Solving this for g, we find:

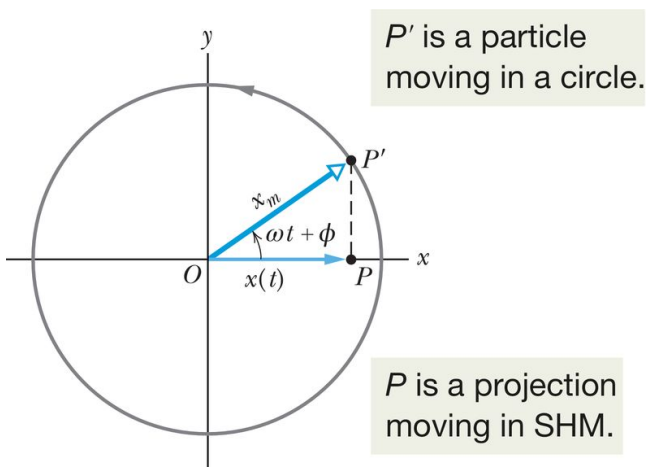
$$g = \frac{8\pi^2 L}{3T^2}$$

This allows experimental determination of g by measuring the period and length of the pendulum.

Simple Harmonic Motion as Circular Motion

Simple harmonic motion (SHM) can also be understood as the projection of **uniform circular motion**.

Imagine a particle moving at constant angular speed ω in a circle.



Its angular position at time t is:

$$\theta(t) = \omega t + \phi$$

If we project the particle's position onto a diameter of the circle (e.g., the x-axis), we get:

$$x(t) = x_m \cos(\omega t + \phi)$$

This is the standard form of simple harmonic motion.

- The **velocity** is:

$$v(t) = -\omega x_m \sin(\omega t + \phi)$$

- The **acceleration** is:

$$a(t) = -\omega^2 x_m \cos(\omega t + \phi) = -\omega^2 x(t)$$

Thus, **circular motion viewed edge-on** results in motion that satisfies the conditions for simple harmonic motion.